

(b) Heat loss to the surroundings.

- 3 (a) (i)** From 0 to 0.1 V, zero current flowing through. The diode needs to be forward biased minimally before conduction of electricity can take place. From 0.2 V to 0.8 V, the current increases rapidly at an increasing rate. The resistance decreases. From 0.8 V onwards, the current increases at constant rate. Resistance stays constant.

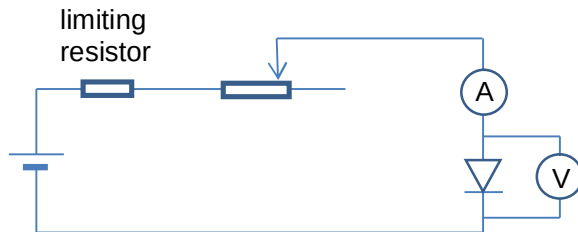
The resistance of the diode was seldom linked to the features of the graph of current against voltage. Many answers described the theory of semiconductors and this was not asked for in the question. A significant number of candidates considered the straight line part of the graph to indicate a constant resistance and also that Ohm's law was obeyed. A large number of candidates incorrectly referred to temperature increase causing more charge carriers and hence less resistance. This answer was not related to the graph or the

semiconductor diode. A significant number referred incorrectly to the inverse of the gradient giving the resistance of the diode.

$$(ii) \text{ Resistance} = \frac{V}{I} = \frac{0.8}{4.4 \times 10^{-3}} = 180 \, \Omega$$

- (b) 1. The minimum forward bias of diode is 0.8 V. At this p.d. across the diode, R needs to be about 160 Ω . This is out of range of R which is between 0 and 100 only.
 2. In practice, it is good to incorporate a limiting resistor in the circuit. If R starts off at 0 Ω , a large current may flow through the diode. Diode may be damaged.

(c) (i)



- (ii) This effectively addresses all the 3 issues in 3(b).

4. (a) Gas molecules move randomly within the cylinder. The molecules collide with the wall